Internet Appendix for
"Late to Recessions: Stocks and the Business Cycle"

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The appendix consists of two main sections. In Section I, I present additional empirical results. In Section II, I provide details on the forecasting model.

I. Additional Empirical Results

Each section below is self-contained and can be skipped without loss.

A. Robustness: Simple strategy

In this section, I provide several robustness checks of the main strategy that trades on macro information. Specifically, in Table I of the main article, given the data limitations, I restricted the analysis to the 1980:M1 through 2019:M12 sample period and used a 12-month moving average to compute the growth/valuation signals. Here, for each series I consider the first observation in which we observe the series, and I consider various moving-average lengths. To compute the increase in Sharpe ratio, I compute the Sharpe ratio of the strategy and divide it by the Sharpe ratio of its fixed allocation

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benchmark computed over the same time period. Importantly, the main results do not change. First, the simple strategy generates larger Sharpe ratios over the fixed allocation benchmark and positive CAPM alphas. Second, the strategy reduces the maximum drawdowns during recessions. Third, macroeconomic variables play an important economic role in identifying economic contractions as measured by the increase in the Sharpe ratio. These results are robust to the moving-average size. For instance, the simple strategy outperforms the buy-and-hold approach when I consider window sizes of length six, 18, or 24 months instead of 12.

Table IA.I.
Performance of Trading Strategies: Robustness

This table reports performance measures for six different strategies. “100% stocks” refers to the buy-and-hold strategy, fully invested in stocks. “76/24 mix” refers to the portfolio that assigns a portfolio weight of 76% to equities and 24% to T-bills. The remaining four strategies switch from stocks to cash during bad times and market downturns while staying fully invested in stocks in the remaining periods. For these strategies, market downturns are months in which returns are below their 12-month moving average. The “Ind. prod.” strategy classifies as a bad time periods in which the industrial production growth series is below its 12-month moving average. The “Common growth” strategy classifies as a bad time periods in which the common growth component of 24 real-time macroeconomic variables is below its 12-month moving average. See Section II in the main text for details on this variable’s construction. The “CAPE” strategy defines as bad times, or expensive periods, periods in which the CAPE ratio is above its 12-month moving average. Finally, the “Slope” strategy classifies as bad times periods in which the yield curve slope is negative. For each strategy, I report the Sharpe ratio increase relative to that obtained from a fixed allocation strategy investing in stocks the same unconditional amount of time. I report t-statistics in brackets below the coefficient estimates and standard errors adjust for heteroskedasticity. I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1,200.

	Fixed allocation		Macro variables		Financial variables	
	100% stocks (1)	76/24 mix (2)	Ind. prod. (3)	Common growth (4)	CAPE (5)	Slope (6)
Time invested	1	0.76	0.76	0.75	0.73	0.93
Mean excess return	7.88	5.32	7.75	7.87	5.66	8.73
S.D.	18.47	12.47	15.26	11.68	16.15	14.28
Sharpe ratio incr.	–	–	18.99	31.80	-17.85	17.78
α-Mkt	–	–	2.38	3.26	-0.32	1.59
	–	–	[2.63]	[2.72]	[-0.38]	[2.22]
Average maximum drawdowns						
Total	17.13	10.72	8.69	6.47	14.52	9.65
Expansions	14.35	8.52	6.96	5.57	12.32	8.55
Recessions	30.44	21.27	16.96	14.02	25.02	19.00
CAPM alpha (α-Mkt) for different sizes of the moving average h						
$h = 6$ months			2.05	3.29	-0.53	1.45
			[2.26]	[2.76]	[-0.64]	[2.01]
$h = 12$ months			2.38	3.26	-0.32	1.59
			[2.63]	[2.72]	[-0.38]	[2.22]
$h = 18$ months			2.62	2.71	-0.77	1.39
			[2.88]	[2.29]	[-0.90]	[1.91]
$h = 24$ months			1.78	3.55	-0.39	1.50
			[1.97]	[2.92]	-[0.46]	[2.08]
Starting Date	1927-06	1927-06	1927-06	1980-12	1927-06	1976-06

B. *Alternative Model Specifications*

In the main article, I assume that the benchmark specification model of returns described in equations (2), (3), and (4) has a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and I allow the fraction of variance in excess returns explained by μ_t to vary across economic regimes, that is, $\phi(S_t)$ can take different values in each state. As discussed in the main text, the estimation of the benchmark specification model points towards a fairly persistent predictable component in realized excess returns with a ρ estimate of 0.97. The estimate of $\rho_{\mu,r}$ is -0.95, which suggests that return surprises and expected return news are strongly negatively correlated. Finally, the estimation shows that discount rates are relatively stable in expansions and much more volatile during recessions.

To assess the robustness of this finding, I now modify the benchmark model along five dimensions:

1. I set $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t.
2. $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.)$.
3. I set $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.)$.
4. I allow $\rho_{\mu,r}$ to vary across states S_t.
5. I allow the persistence parameter ρ to vary across states S_t.

The parameter estimates of these five alternative models are presented in Table IA.II. At the bottom, I also report the log marginal data density, which provides the formal test in terms of model fit. Note that the difference in $\ln p(Y)$, gives log posterior odds when the prior odds are equal to one.

Several aspects of these estimation results are worth noting. First, based on $\ln p(Y)$, I conclude that the benchmark specification is preferred over these model alternatives. Second, when $\rho_{\mu,r} = 0$ or $\phi(S_t = Exp.) = \phi(S_t = Rec.)$, the log marginal data density, $\ln p(Y)$, drops significantly, which implies that a strong feature in the data is $\rho_{\mu,r} < 0$ (see Specification 1) and that discount rates are fairly stable in expansions and much more volatile in recessions (see Specification 2).

Third, in Specification 4 I allow $\rho_{\mu,r}(S_t)$ to depend on the state S_t. Note that the 90% confidence intervals for $\rho_{\mu,r}(S_t = Exp.)$ and $\rho_{\mu,r}(S_t = Rec.)$ are remarkably similar. Using the posterior parameter estimates, I formally test the null hypothesis

$$H_0 : \rho_{\mu,r}(S_t = Exp.) = \rho_{\mu,r}(S_t = Rec.), \tag{IA.1}$$

which I cannot reject at standard levels of significance. The 90% confidence interval on the difference equals $(-0.008, 0.017)$. Economically, this result suggests that the fraction of variance in return news explained by discount rate revisions does not vary across economic regimes.

Fourth, in Specification 5 I allow $\rho(S_t)$ to depend on the state S_t. The main results here are that the estimate of ρ is approximately 0.98 in expansions, while it is equal to 0.95 in recessions. I test whether the persistence coefficient of expected returns is higher in expansions via the null hypothesis

$$H_0 : \rho(S_t = Rec.) \leq \rho(S_t = Exp.), \tag{IA.2}$$

which I cannot reject at standard levels of significance. The 90% confidence interval on the difference (i.e., $\rho(S_t = Exp.) - \rho(S_t = Rec.)$) equals $(0.027, 0.034)$. These results suggest that risk premia shocks die out more quickly during recessions. For instance, based on the posterior median estimates, the half-life of a discount rate shock is 34 months in expansions and 13 months in recessions.[1] However, one should take care in drawing conclusions based on this analysis because this model is dominated, in terms of model fit, by the benchmark specifications since the log odds equals 10.

Model-Implied Autocorrelation and Kalman Gains. Next, I compute the model-implied autocorrelation of returns and the Kalman gains. In the main article, I showed that, conditioning on each economic regime, the autocorrelation of returns is given by

$$corr_{S_t}(r^e_{t+1}, r^e_t) = \rho(S_t) \cdot \phi(S_t) + \rho_{\mu,r}(S_t) \cdot \sqrt{\phi(S_t)(1-\phi(S_t))(1-\rho(S_t)^2)}. \tag{IA.3}$$

For each model, I evaluate (IA.3) using parameter draws from their posterior distribution. Table IA.III reports the 5$^{\text{th}}$, 50$^{\text{th}}$, and 95$^{\text{th}}$ percentiles from this computation.[2] Table IA.III shows that the main model generates a strong

[1] The half-life is defined as the number of months t required for the risk premia shock to be reduced to half, that is, $\rho^t = 1/2$.

[2] Specifically, I sampled 10,000 draws of the model parameters from their posterior

positive autocorrelation in returns during recessions and mild negative autocorrelation in returns during expansions, with values of around 6.4% and -1.2%, respectively. If I set $\rho_{\mu,r} = 0$, then the autocorrelation in returns during expansions turns positive and is equal to 1.2% (see Specification 1). Reverting back to $\rho_{\mu,r} < 0$ and setting $\phi(S_t = Exp.) = \phi(S_t = Rec.)$ produces a return autocorrelation in expansions and in both recessions that is small and negative with values of around -1.2 (see Specification (2)). The combination of $\rho_{\mu,r} = 0$ and $\phi(S_t = Exp.) = \phi(S_t = Rec.)$ implies a mild positive autocorrelation in returns throughout (see Specification 3). Allowing $\rho_{\mu,r}$ to depend on S_t also produces a state-dependent pattern in the autocorrelation in returns, that is, strong positive autocorrelation in recessions and mild reversals during expansions (see Specification 4). This result is not surprising given that I could not reject the null hypothesis in (IA.1). Finally, when ρ depends on S_t, the model-implied autocorrelation in returns is state dependent, but the serial correlation in returns during recessions drops substantially from 6.4% to 2.7% given that the persistence in expected returns drops during recessions (see Specification 5).

Table IA.III also reports the 5^{th}, 50^{th}, and 95^{th} percentiles for the steady-state Kalman gain using parameter draws from their posterior distribution. As explained in the main text, the Kalman gain determines the weight given to past returns when computing the estimate of expected returns. The Kalman gains across model specifications mimic the model-implied state-

distribution. I then evaluated equation (IA.3) at those values, which for each regime and each model gives 10,000 different values of the model-implied autocorrelation in returns. Finally, I computed the 5^{th}, 50^{th}, and 95^{th} percentiles.

dependent autocorrelations in returns.

Matching the State-Dependent Autocorrelation in Returns. I now examine the extent to which these alternative model specifications account for the state-dependent autocorrelation in returns as implied by the OLS regression coefficients in Table II of the main article. To do so, for each model, I compute this predictive check as follows. First, I draw 10,000 parameters from the posterior distribution. For each parameter draw, I simulate trajectories for returns and the common growth component (described in Section II). Both of these trajectories have the same number of observations as in the data (660 monthly observations). I then compute the model-implied regression coefficients by running the specification

$$r^e_{t+1} = a_0 + \left(b_0 + b_{Rec} \cdot I_{\text{Comm.Growth},t}\right) r^e_t + \epsilon_{t+1}, \tag{IA.4}$$

where $I_{\text{Comm.Growth},t}$ takes a value of one whenever the simulated common growth component is low relative to its 12-month moving average and zero otherwise.

Results are summarized in Table IA.IV. I focus on the results from estimating the benchmark specification. All three OLS regression point estimates implied by the actual data are within the 5^{th} and 95^{th} percentile of the corresponding posterior predictive distribution. In particular, the benchmark model generates a sizeable return autocorrelation conditional on a recession, as implied by the regression coefficients, with a median value of 0.11 ($=\hat{b}_0+\hat{b}_{Rec}$), which is remarkably close to point estimates from the data equal to 0.09. Moreover, consistent with the data, in good economic times the autocorrelation in returns as implied by (IA.4) is -0.01, but not statistically significant.

The five alternative model specifications are not able to capture this feature of the data. The only exception is Specification 4 since, in essence, the posterior estimates are very close to the benchmark specification (see Table IA.II).

I assess the economic relevance of these model ingredients in Section F below.

Table IA.II.

Alternative Model Specifications: Posterior Estimates

This table reports the 5th, 50th, and 95th percentiles of the posterior distribution for six different model specifications. The benchmark specification model of returns assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 1, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t. In Specification 2, $\rho_{\mu,r}$ is a free parameter to be estimated and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 4, I allow $\rho_{\mu,r}$ to vary across states S_t. In Specification 5, I allow the persistence parameter ρ to vary across states S_t. The estimation sample runs from January 1965 to December 2019.

	Benchmark specification				Specification 1				Specification 2		
					$\rho_{\mu,r} = 0$				$\phi(S_t) = \phi$		
	5%	50%	95%		5%	50%	95%		5%	50%	95%
μ_0	0.005	0.006	0.007	μ_0	0.005	0.006	0.007	μ_0	0.005	0.006	0.007
ρ	0.958	0.970	0.982	ρ	0.958	0.970	0.982	ρ	0.958	0.970	0.982
$\rho_{\mu,r}$	-0.984	-0.955	-0.926	$\rho_{\mu,r}$	0.000	0.000	0.000	$\rho_{\mu,r}$	-0.950	-0.924	-0.893
$\phi(S_t = Rec.)$	0.093	0.149	0.181	$\phi(S_t = Rec.)$	0.010	0.145	0.171	ϕ	0.010	0.012	0.014
$\phi(S_t = Exp.)$	0.008	0.009	0.011	$\phi(S_t = Exp.)$	0.012	0.013	0.014		–	–	–
$\sqrt{Var(r_t^e \| S_t = Rec.)}$	0.043	0.060	0.072	$\sqrt{Var(r_t^e \| S_t = Rec.)}$	0.045	0.058	0.069	$\sqrt{Var(r_t^e \| S_t = Rec.)}$	0.047	0.058	0.069
$\sqrt{Var(r_t^e \| S_t = Exp.)}$	0.039	0.039	0.039	$\sqrt{Var(r_t^e \| S_t = Exp.)}$	0.039	0.039	0.039	$\sqrt{Var(r_t^e \| S_t = Exp.)}$	0.039	0.039	0.039
$\ln p(Y)$		1475		$\ln p(Y)$		1286		$\ln p(Y)$		1473	
	Specification 3				Specification 4				Specification 5		
	$\rho_{\mu,r} = 0$ & $\phi(S_t) = \phi$				$\rho_{\mu,r}(S_t)$				$\rho(S_t)$		
	5%	50%	95%		5%	50%	95%		5%	50%	95%
μ_0	0.005	0.006	0.007	μ_0	0.005	0.006	0.007	μ_0	0.005	0.006	0.007
ρ	0.958	0.970	0.982	ρ	0.958	0.970	0.981	$\rho(S_t = Rec.)$	0.948	0.950	0.951
	–	–	–		–	–	–	$\rho(S_t = Exp.)$	0.979	0.980	0.982
$\rho_{\mu,r}$	0.000	0.000	0.000	$\rho_{\mu,r}(S_t = Rec.)$	-0.973	-0.949	-0.938	$\rho_{\mu,r}$	-0.976	-0.951	-0.916
	–	–	–	$\rho_{\mu,r}(S_t = Exp.)$	-0.979	-0.952	-0.921		–	–	–
ϕ	0.011	0.015	0.018	$\phi(S_t = Rec.)$	0.112	0.143	0.169	$\phi(S_t = Rec.)$	0.121	0.135	0.152
	–	–	–	$\phi(S_t = Exp.)$	0.005	0.007	0.011	$\phi(S_t = Exp.)$	0.006	0.007	0.014
$\sqrt{Var(r_t^e \| S_t = Rec.)}$	0.048	0.061	0.073	$\sqrt{Var(r_t^e \| S_t = Rec.)}$	0.049	0.063	0.075	$\sqrt{Var(r_t^e \| S_t = Rec.)}$	0.042	0.056	0.068
$\sqrt{Var(r_t^e \| S_t = Exp.)}$	0.039	0.039	0.039	$\sqrt{Var(r_t^e \| S_t = Exp.)}$	0.039	0.039	0.039	$\sqrt{Var(r_t^e \| S_t = Exp.)}$	0.039	0.039	0.039
$\ln p(Y)$		1283		$\ln p(Y)$		1468		$\ln p(Y)$		1465	

Table IA.III.

Alternative Model Specifications: Model-Implied Serial Correlation and Kalman Gains

This table reports the 5^{th}, 50^{th}, and 95^{th} percentiles for the model-implied autocorrelation in returns and the Kalman gain for six different model specifications. To compute these statistics, I sample model parameters from their posterior distribution. The benchmark specification model of returns assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 1, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t. In Specification 2, $\rho_{\mu,r}$ is a free parameter to be estimated and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 4, I allow $\rho_{\mu,r}$ to vary across states S_t. In Specification 5, I allow the persistence parameter ρ to vary across states S_t. The estimation sample runs from January 1965 to December 2019.

	Benchmark specification			Specification 1 $\rho_{\mu,r} = 0$			Specification 2 $\phi(S_t) = \phi$		
	5%	50%	95%	5%	50%	95%	5%	50%	95%
				Model-implied serial correlation in returns					
Expansion	-1.320	-1.190	-1.081	1.129	1.251	1.413	-1.391	-1.258	-1.145
Recession	5.919	6.457	6.872	13.262	14.035	14.637	-1.391	-1.258	-1.145
				Kalman gains					
Expansion	-0.013	-0.012	-0.011	0.009	0.012	0.014	-0.014	-0.013	-0.011
Recession	0.058	0.064	0.068	0.133	0.140	0.146	-0.014	-0.013	-0.011

	Specification 3 $\rho_{\mu,r} = 0$ & $\phi(S_t) = \phi$			Specification 4 $\rho_{\mu,r}(S_t)$			Specification 5 $\rho(S_t)$		
	5%	50%	95%	5%	50%	95%	5%	50%	95%
				Model-implied serial correlation in returns					
Expansion	1.154	1.446	1.737	-1.418	-1.280	-1.147	-0.963	-0.872	-0.790
Recession	1.154	1.446	1.737	4.481	5.373	6.031	2.384	2.721	3.264
				Kalman gains					
Expansion	0.009	0.013	0.017	-0.014	-0.013	-0.012	-0.010	-0.009	-0.008
Recession	0.009	0.013	0.017	0.045	0.055	0.060	0.024	0.028	0.033

Table IA.IV.
Alternative Model Specifications: Regression Evidence on the Serial Correlation in Returns

This table reports the 5^{th}, 50^{th}, and 95^{th} percentiles for the model-implied regression coefficients. Using model-simulated data, I run the specification $r^e_{t+1} = a^j_0 + \left(b^j_0 + b^j_1 \cdot I_{\text{Comm.Growth},t}\right) r^e_t + \epsilon_{t+1}$, where $I_{\text{Comm.Growth},t}$ takes a value of one when the simulated common growth component is low relative to its $12-$month moving average and zero otherwise. The simulated data have the same number of observations as in the actual sample. I consider six model specifications. The benchmark specification model of returns assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 1, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t. In Specification 2, $\rho_{\mu,r}$ is a free parameter to be estimated and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 4, I allow $\rho_{\mu,r}$ to vary across states S_t. In Specification 5, I allow the persistence parameter ρ to vary across states S_t. The estimation sample runs from January 1965 to December 2019.

	OLS	Benchmark specification			Specification 1 $\rho_{\mu,r}=0$			Specification 2 $\phi(S_t)=\phi$		
	Est.	5%	50%	95%	5%	50%	95%	5%	50%	95%
$\hat{a}_0$	7.24	2.23	7.40	12.72	-0.52	7.06	14.62	6.49	7.84	9.20
$\hat{b}_0$	-0.01	-0.08	-0.01	0.05	-0.01	0.07	0.14	-0.08	-0.02	0.05
$\hat{b}_{\text{Rec}}$	0.10	0.01	0.12	0.23	-0.06	0.05	0.16	-0.10	0.01	0.12
	OLS	**Specification 3 $\rho_{\mu,r}=0$ & $\phi(S_t)=\phi$**			**Specification 4 $\rho_{\mu,r}(S_t)$**			**Specification 5 $\rho(S_t)$**		
	Est.	5%	50%	95%	5%	50%	95%	5%	50%	95%
$\hat{a}_0$	7.24	3.34	7.56	11.88	2.41	7.45	12.49	2.53	7.64	12.71
$\hat{b}_0$	-0.01	-0.06	0.01	0.07	-0.09	-0.02	0.05	-0.07	-0.01	0.06
$\hat{b}_{\text{Rec}}$	0.10	-0.10	0.01	0.12	0.02	0.13	0.24	-0.06	0.05	0.16

C. Considering Different Pricing Anomalies and Subsample Periods

Table IA.V shows that the main specification has a positive alpha with respect to a wide range of pricing factors and across different subsample periods. I present results for several combinations of factors including the Fama-French three factors, the momentum factor, the time-series momentum factor, and the volatility-managed factor.

Reassuringly, for the full 1980 to 2019 sample period, all alphas are statistically significant and economically large, ranging from 3.91% to 6.12%. One might worry that these results are driven entirely by the Great Recession that began in December 2007 and ended in June 2009. To address this concern, I also analyze the performance of the strategy across two subsamples ranging from 1980 to 2000 and from 2001 to 2019. In both subsamples, the alphas are economically large and statistically significant.

D. Leverage, Short-Selling, and Transaction Costs

I next show that the strategy produces fairly large benefits even with tight leverage and short-selling constraints. Moreover, transaction costs would need to be incredibly high to make the strategy unprofitable.

Panel A of Table IA.VI reports the distribution of the portfolio weights under different leverage scenarios and short-selling constraints together with various performance measures. The first row is the benchmark strategy, which does not impose any constraints. Notably, the benchmark strategy uses modest levels of short-selling and leverage: the 5% and 95% quintiles

Table IA.V.
Multi-Factor Alphas and Subsample Analysis

This table reports alphas from time-series regressions of the main strategy returns on different asset pricing factors. With respect to the pricing factors, I consider the Fama-French three factors, the momentum factor (Mom), the time-series momentum factor (TMom), and the volatility-managed factor (TVol). I also report results for subsample regressions. Standard errors adjust for heteroskedasticity and autocorrelation. I report t-statistics in brackets. To facilitate interpretation, all returns are annualized in percent per year by multiplying monthly returns by 1,200. The estimation sample runs from January 1980 to December 2019.

	Mkt (1)	Mkt TMom (2)	Mkt TVol (3)	Mkt TMom TVol (4)	FF3 (5)	FF3 Mom (6)	FF3 TMom TVol Mom (7)
α: 1980-2019	5.38	4.46	4.33	3.91	6.12	4.94	4.67
	[2.76]	[3.41]	[3.19]	[3.69]	[2.50]	[2.08]	[2.95]
α: 1980-2000	4.63	4.57	3.88	3.82	4.43	4.46	3.61
	[2.82]	[2.66]	[2.15]	[1.99]	[2.15]	[2.32]	[1.90]
α: 2001-2019	6.95	3.05	5.98	3.11	6.93	6.25	3.40
	[2.49]	[2.63]	[3.37]	[2.56]	[2.59]	[2.28]	[3.77]

of the weight distribution are -0.31 and 1.52, respectively.[3] These numbers suggest that the performance of the main strategy is not due to unrealistically high levels of short-selling or leverage. To see this, in the second row I preclude the strategy from going short, while in the third and fourth rows I add a leverage constraint by imposing a cap on the upper bound of the portfolio weight of 1.5 (50% standard margin requirement) and 1.0 (no leverage), respectively. Several aspects of these restrictions are worth noting here, I simply mention a few. First, these constraints substantially reduce trading activity as measured by the mean absolute difference of the monthly

[3]The 1% and 99% quintiles are -1.45 and 1.98, respectively.

Table IA.VI.
Leverage, Short-Selling, and Transaction Costs

This table reports various performance measures for several alternative constraints on the main strategy and for different transaction costs. The alternative constraints consist of restricting the risk exposure to be above zero (i.e., no short-selling), below 1 (i.e., no leverage), or below 1.5 (50% standard margin requirement). With respect to the transaction costs, I consider 1, 10, and 20 basis points. Standard errors adjust for heteroskedasticity and autocorrelation. I report t-statistics in brackets. To facilitate an interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1,200. All models are estimated out of sample on recursively increasing samples of data. The forecast evaluation period runs from January 1980 to December 2019.

	Panel A: Leverage and short-selling constraints								Panel B: Transaction costs			
weight		Distribution of Weights ω_t						Sharpe	α for different transaction costs			
limits	$\|\Delta\omega\|$	5%	25%	50%	75%	95%	$E[R]$	increase	None	1bps	10bps	20bps
[− , −]	0.23	-0.31	0.56	0.93	1.15	1.52	10.31	28.32	5.38	5.41	5.15	4.87
									[2.35]	[2.36]	[2.27]	[2.17]
[0, −]	0.20	0.00	0.56	0.93	1.15	1.52	9.13	28.28	3.19	3.21	3.00	2.76
									[2.77]	[2.81]	[2.64]	[2.44]
[0, 1.5]	0.18	0.00	0.56	0.93	1.15	1.50	9.09	30.28	3.22	3.23	3.04	2.82
									[2.93]	[2.95]	[2.79]	[2.61]
[0, 1.0]	0.11	0.00	0.56	0.93	1.00	1.00	7.78	31.73	2.63	2.64	2.52	2.38
									[3.17]	[3.18]	[3.05]	[2.90]

weights, $|\Delta\omega|$. However, the overall performance of the strategies does not decrease much. For instance, even under the extreme case of no short-selling and leverage, the strategy earns a significant alpha of 2.63% and increases Sharpe ratios relative to the buy-and-hold approach by 31%.

Panel B of Table IA.VI shows that the performance of the strategy survives realistic levels of transaction costs. I report the estimated CAPM alpha under various trading costs: 1, 10, and 20 basis points. To put these numbers into perspective, Fleming et al. (2003) estimate a cost of around 1 basis point of the traded value in the futures market. Overall, the results of Panel B suggest that transaction costs would need to be incredibly high to eliminate the market-timing gains.

E. Additional Market-Timing Tests

In this section, I present five additional timing-related performance measures. Column (1) of Table IA.VII presents the results. The first measure attempts to capture the trade-off between risk and returns via a utility function. Following Fleming et al. (2001), I equate the realized quadratic utility obtained under the dynamic portfolio strategy with the one obtained under the static buy-and-hold strategy,

$$\sum_{t=0}^{T-1}\left[\left(r_{t+1}^{e,\mu}-\Delta\right)-\frac{\gamma}{2(1+\gamma)}\left(r_{t+1}^{e,\mu}-\Delta\right)^2\right]=\sum_{t=0}^{T-1}\left[\left(r_{t+1}^{e}\right)-\frac{\gamma}{2(1+\gamma)}\left(r_{t+1}^{e}\right)^2\right], \tag{IA.5}$$

and solve for Δ for a given level of risk aversion γ.[4] The parameter Δ can be interpreted as the monthly maximum performance fee that an investor would be willing to pay to have access to the dynamic strategy. As shown in Panel A, the performance fee for the main strategy is substantial. For instance, an investor with a quadratic utility and a risk aversion coefficient of three is willing to pay a monthly fee of around 19 basis points to have access to the

[4]More generally, the quadratic utility function can be viewed as a second-order approximation to the investor's utility function. I assume a risk aversion coefficient equal to three to compute the performance fee. However, I verified that the outperformance is robust to different perturbations to γ.

$\hat{\mu}_{t+1|t}$ estimate.

The second market-timing test is the manipulation-proof performance measure (MPPM) of Goetzmann et al. (2007). Traditional measures (e.g., alphas, appraisal ratios, Sharpe ratios) are vulnerable because they can be manipulated.[5] The MPPM statistic is robust to manipulations and can be written as

$$MPPM = \frac{1}{(1-\rho)\Delta t} \ln \left(\frac{1}{T} \sum_{t=1}^{T} \left[\frac{1+r_{p,t}}{1+r_{f,t}} \right]^{1-\rho} \right), \tag{IA.6}$$

where $r_{f,t}$ denotes the risk-free rate and $r_{p,t}$ denotes the per-period (not annualized) portfolio return. I select the coefficient ρ ($\approx$ 2.7) such that holding the market is optimal for an uninformed manager.[6] Panel A shows that the MPPM statistic for the main specification is equal to 7.8% and can be interpreted as the annualized continuously compounded excess return certainty equivalent of the portfolio.

The third market-timing test is straightforward. A successful market-timing investor would increase their exposure to the risky asset when future returns are high and would decrease it when future returns are low. Hence, the correlation between the forecasting weights $\hat{\mu}_{t+1|t}$ and the time $t+1$ realized returns, $r^e_{m,t+1}$, should be positive. Indeed, the last row of Panel A

[5] Portfolio managers can use dynamic strategies or derivatives to alter the distribution of returns to artificially enhance any particular performance measure or deliberately generate measurement errors to their benefit. For details and examples, see Goetzmann et al. (2007).

[6] Specifically, I set $\rho = \frac{\ln[E(1+\hat{r}_m)-\ln(1+\hat{r}_f)]}{Var(1+\hat{r}_m)}$, where $\hat{r}_m$ and $\hat{r}_f$ are the unconditional mean of the market and risk-free rate.

shows a positive correlation of 13% for the main specification.

The fourth measure is based on the quadratic regression of Treynor and Mazuy (1966),

$$r_{p,t+1} = \gamma_0 + \gamma_1 r^e_{m,t+1} + \gamma_2 \left(r^e_{m,t+1}\right)^2 + \epsilon_{t+1}, \tag{IA.7}$$

where the coefficient γ_2 measures return timing performance. As shown in Panel B of Table IA.VII, the γ_2 coefficient associated with the main predictor is significantly positive. As expected, given that the quadratic regression term $\left(r^e_{m,t+1}\right)^2$ captures timing ability, the abnormal return γ_0 for the main specification is small and insignificant.[7]

The fifth measure is due to Henriksson and Merton (1981) and replaces the quadratic term $\left(r^e_{m,t+1}\right)^2$ with $\max(-r^e_{m,t+1}, 0)$,

$$r_{p,t+1} = \gamma_0 + \gamma_1 r^e_{m,t+1} + \gamma_2 \max(-r^e_{m,t+1}, 0) + \epsilon_{t+1}. \tag{IA.8}$$

A positive γ_2 suggests market-timing ability. Relative to the measure of Treynor and Mazuy (1966), the emphasis of this market-timing measure is on how well it predicts during market downturns. Panel C of Table IA.VII reports the results for this specification. The positive γ_2 under the main specification provides further evidence in support of market-timing ability during downturns.

[7]After controlling for $\left(r^e_{m,t+1}\right)^2$, a positive γ_0 would suggest superior stock selection, which is not considered in this paper.

Table IA.VII.
Additional Market-Timing Tests

This table presents five market-timing tests for the different specifications considered in the main text. Panel A shows a utility-based performance measure (Performance fee), the manipulation-proof performance measure (MPPM) of Goetzmann et al. (2007), and the correlation between the portfolio weights at time t and the $t+1$ realized excess returns ($Corr(\hat{\mu}_{t+1|t}, r^e_{m,t+1})$). Panels B and C show results for the timing measures of Treynor and Mazuy (1966) and Henriksson and Merton (1981). Column (1) presents results for the benchmark specification model of returns, which assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In specification (2), I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t. In specification (3), I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In specification (4), $\rho_{\mu,r}$ is a free parameter to be estimated and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. I report t-statistics in brackets. All models are estimated out of sample on recursively increasing samples of data. The forecast evaluation period runs from January 1980 to December 2019.

		Model Constraints		
	Main (1)	$\rho_{\mu,r} = 0$ (2)	$\rho_{\mu,r} = 0$ $\phi(S_t) = \phi$ (3)	$\phi(S_t) = \phi$ (4)
Panel A: Performance measures				
Performance fee (bps)	18.95	-9.91	-0.60	-3.49
MPPM	7.79	4.56	4.78	4.10
$Corr(\hat{\mu}_{t+1\|t}, r^e_{m,t+1})$	0.13	0.04	0.06	-0.05
Panel B: Results for the timing measures of Treynor and Mazuy (1966)				
Regression: $r_{p,t+1} = \gamma_0 + \gamma_1 r^e_{m,t+1} + \gamma_2 \left(r^e_{m,t+1}\right)^2 + \epsilon_{t+1}$				
γ_0	-0.01	3.88	-2.47	2.02
	[-0.00]	[1.74]	[-2.34]	[2.68]
γ_1	0.65	0.47	0.91	0.94
	[4.79]	[2.31]	[8.31]	[13.08]
γ_2	2.16	-0.33	1.30	-0.82
	[1.77]	[-0.28]	[2.97]	[-2.03]
Panel C: Results for the timing measures of Henriksson and Merton (1981)				
Regression: $r_{p,t+1} = \gamma_0 + \gamma_1 r^e_{m,t+1} + \gamma_2 max(-r^e_{m,t+1}, 0) + \epsilon_{t+1}$				
γ_0	-5.44	2.36	-4.46	2.78
	[-1.77]	[0.68]	[-2.45]	[1.95]
γ_1	0.88	0.49	1.02	0.88
	[8.39]	[2.82]	[8.22]	[12.30]
γ_2	0.52	0.03	0.25	-0.14
	[2.37]	[0.14]	2[.28]	[-1.45]

Table IA.VIII.
Economic Importance of Model Ingredients

This table presents results of the trading strategy under various forecasting model constraints. Column (1) presents results for the benchmark specification model of returns, which assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 2, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t. In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 4, $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. Standard errors adjust for heteroskedasticity and autocorrelation. I report t-statistics in brackets. All models are estimated out of sample on recursively increasing samples of data. The forecast evaluation period runs from January 1980 to December 2019.

		Model Constraints		
	Main (1)	$\rho_{\mu,r} = 0$ (2)	$\rho_{\mu,r} = 0$ $\phi(S_t) = \phi$ (3)	$\phi(S_t) = \phi$ (4)
α - Mkt	5.46	3.05	0.82	-0.05
	[2.77]	[1.58]	[1.03]	[-0.07]
Sharpe ratio increase $\times 100\%$	28.23	-14.87	-0.96	-5.27
R^2_{OOS}	1.671	-0.126	0.493	-0.232
R^2_{OOS} (p-values)	(0.026)	(0.304)	(0.100)	(0.697)

F. Importance of the Two Key Model Ingredients.

To assess the economic relevance of the two key assumed model ingredients, I turn each one of them off separately and re–estimate the model. Table IA.VIII presents the results. Column (1) presents results for the main specification, which assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In column (2), I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t.

In column (3), I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In column (4), $\rho_{\mu,r}$ is a free parameter to be estimated and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$.

The main takeaway is that the profitability of the strategy disappears when I impose the model constraints. The market alphas are small and statistically insignificant. The Sharpe ratios are lower than the Sharpe ratios of the buy-and-hold strategy. The R^2_{OOS} statistics are negative. These results stress the economic importance of the two key model ingredients and complement the results in Section B above, where I assess statistical significance by computing the log marginal data density, $\ln p(Y)$, for each specification.

G. *Characteristic-Sorted Portfolio Returns*

Table IA.IX presents the results when, instead of timing the excess market return, I time characteristic-sorted portfolio returns in excess of the risk-free rate. Specifically, I consider the first, third, and fifth quintile on stock book-to-market ratios, market capitalization, and momentum portfolios from Kenneth R. French's website.[8] I report results only for the main specification. As expected, the strategy works equally well for these returns. The strategy produces positive alphas with respect to the three Fama-French factors (Fama and French, 1992) and the momentum factors and it increases Sharpe ratios. Finally, the last row reports the out-of-sample R^2 above 1% per month.

[8]See `https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`.

Table IA.IX.
Characteristic-Sorted Portfolio Returns

This table presents results of the economic and statistical significance of the characteristic-sorted portfolio excess returns. I focus on the results for the main specification estimated out of sample using an expanding window. Standard errors adjust for heteroskedasticity and autocorrelation. I report t-statistics in brackets. To facilitate interpretation, all returns are annualized in percent per year by multiplying monthly returns by 1,200. The forecast evaluation period runs from January 1980 to December 2016.

	Growth to Value			Big to Small			Losers to Winners		
	(1)	(3)	(5)	(1)	(3)	(5)	(1)	(3)	(5)
Panel (a): Market -timing $\omega_t \propto E_t[r_{t+1}]$									
alpha - FF3 Mom	5.67	5.39	4.91	6.28	4.08	2.36	8.25	4.29	4.41
	[2.69]	[2.97]	[2.27]	[3.21]	[2.68]	[1.46]	[1.99]	[2.53]	[2.51]
Original Sharpe	0.53	0.64	0.59	0.59	0.53	0.42	0.13	0.55	0.64
New Sharpe	0.68	0.83	0.75	0.79	0.66	0.50	0.21	0.67	0.83
$R^2_{OOS}(\%)$	1.38**	2.09***	2.20***	1.88***	2.12***	1.89***	3.29***	2.17***	1.04*

H. Extended Sample Period

Above, I report out-of-sample results based on a 1980:01 to 2019:12 sample split. In this section, I demonstrate the robustness of the out-of-sample forecasts to an earlier sample period. Ideally, I should predict economic states recursively using real-time data as in the main text. However, most of the macro data available to use start in 1960. Thus, the earliest that I can start out-of-sample forecasts using this approach is from 1980 onwards (given that I need approximately 20 years of data as a training sample).

To address this issue, I employ the ex post NBER recession indicator. This allows me to extend the forecasting sample period to as early as 1960, which uses around 30 years of data as the training sample (from 1930:01 to 1959:12). With this in mind, I modify the aggregate predictive density in

Table IA.X.
Performance Evaluation: Extended Period

This table presents results after introducing stale information related to economic conditions to the forecasting strategy. The columns correspond to the induced lag in months in the NBER recession dummy. Standard errors adjust for heteroskedasticity and autocorrelation. I report t-statistics in brackets. The sample runs from January 1960 to December 2016. The training sample runs from January 1930 to December 1959.

	Lag in the forecasting weights relative to the NBER Recessionary Index							
	0	1	2	3	$\cdots$	6	9	12
Time Period 1960 - 2016								
alpha - Mkt	6.10	5.16	5.35	5.30		4.38	1.48	0.79
	[3.20]	[2.73]	[2.83]	[2.81]		[2.36]	[0.85]	[0.51]
Original Sharpe	0.33	0.33	0.33	0.33		0.33	0.33	0.33
New Sharpe	0.52	0.46	0.48	0.47		0.43	0.27	0.27
$R^2_{OOS}(\%)$	1.94***	1.48***	1.60***	1.59***		1.11**	-0.41	-0.52

equation (5) in the main text as follows:

$$p(r^e_{t+1}|r^e_{1:t}, \Theta, \pi_t) = (1 - I_{rec,t-h})\ p(r^e_{t+1}|r^e_{1:t}, \Theta_{S_t=0}, S_t = 0) + I_{rec,t-h}\ p(r^e_{t+1}|r^e_{1:t}, \Theta_{S_t=1}, S_t = 1), \tag{IA.9}$$

where $I_{rec,t-h}$ is the NBER recession indicator, which takes a value of one if period $t-h$ is recessionary and zero if it is expansionary. Importantly, I introduce stale information about the economic state by lagging the recessionary indicator by h periods at the time of the forecasts and form portfolios as before. This exercise captures an investor needing h periods to identify that they are in a recession. For instance, when $h = 0$, the investor perfectly foresees recessions and expansions, while $h = 3$ means that they need around three months of data to identify business–cycle turning points.

Table IA.X reports the results. I draw two important conclusions from these results. First, the strategy remains profitable even under stale information about the cycle. However, the outperformance of the strategy vanishes after a six-month lag. Note that for values of h greater than six, the alphas are insignificant and small. The R^2_{00S} statistics are negative. Second, there is strong evidence that the strategy works well for the extended sample period.

I. Stocks and Business Cycles in U.S. States

In this section I extend the sample by looking at the cross-section of U.S. state portfolios and regional business cycles. The motivation for this exercise comes, first, from finance literature underscoring the significance of geographic location in financial decisions. For example, Pirinsky and Wang (2006) find that stock returns of firms headquartered in the same location exhibit substantial comovement, whereas Korniotis and Kumar (2013) suggest that changes in local macroeconomic conditions influence returns on local stocks. In addition, the macroeconomic literature shows that state-level business–cycle phases are not in sync with national phases.[9] For instance, Owyang et al. (2005) and Hamilton and Owyang (2012) find that state recessions differ significantly in the timing of business–cycle turning points, with some states entering or recovering from recessions before others. Therefore, the cross-section of U.S states provides additional data to assess the robust-

[9]References on geographic segmentation in financial markets include: Coval and Moskowitz (1999), Pirinsky and Wang (2006), Hong et al. (2008), and Korniotis and Kumar (2013). Papers that characterize regional business cycles include: Crone (2005), Owyang et al. (2005), Owyang et al. (2008), and Hamilton and Owyang (2012).

ness of my main result. This robustness check leads to consistent results.[10]

I.1. State-level Database and Estimation

To construct value-weighted state portfolios, I obtain stock returns from the Center of Research in Security Prices (CRSP). I restrict the sample to NYSE, Amex, or NASDAQ firms with common equity (i.e., CRSP share code 10 and 11). To form state portfolios, I use the location of the firms' headquarters, consistent with Coval and Moskowitz (1999), Pirinsky and Wang (2006), Hong et al. (2008), and Korniotis and Kumar (2013). I obtain the location data from Compustat. I then cross-check, on an annual basis, the historic records of firms' location from the Compact Disclosure database. This allows me to identify those firms that changed the state of their headquarters in any given year. Finally, I exclude states with less than 10 firms.

To construct the state-level business cycle phases, I use six state-level macroeconomic indicators that are available at a monthly or quarterly frequency. The monthly variables are nonagricultural payroll employment (first available observation: 1990.01), unemployment rate (1976.01), and initial jobless claims (1986.02). The quarterly variables are real wage and salary disbursements (1998.03), personal income (1948.03), and a state-level house

[10]Hong et al. (2007) show that industry portfolios are tightly linked to economic indicators. Specifically, they find that industry returns forecast various indicators of economic activity. Hence, alternatively, I could have used industry portfolios and constructed industry-level business cycles. However, one of the significant hurdles in constructing business–cycle phases in U.S. industries is data availability. Most of the industry macro data are at the annual frequency.

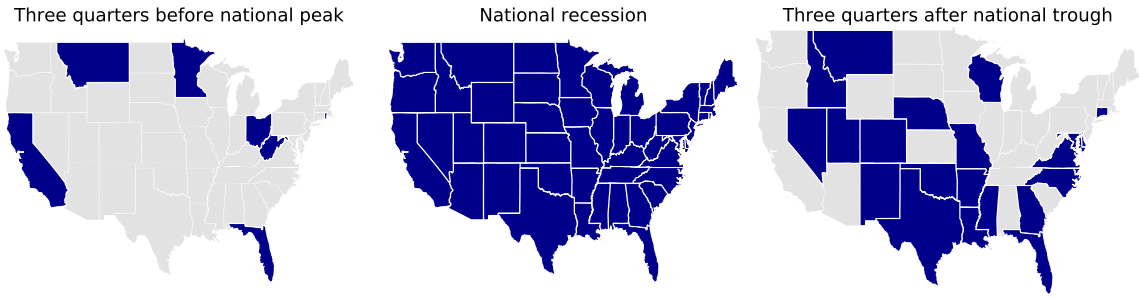

Figure IA.1. National recession 2007 to 2009. This figure shows the diffusion of the 2007 to 2009 national recession across the 48 contiguous states. A U.S. state is taken to be in a recession whenever its recession probability is above 50%.

price index (1975.03). In estimating the economic regimes, I incorporate the series when they become available. As described in Section II.C of the main article, the state-space model is flexible enough to handle mixed-frequency data and to account for missing data coming from the different publication lags.

I.2. State-level Results

For each state, I fit the model of expected returns described in Section II. Given data availability, I start the out-of-sample computations in January 1990 and end in December 2019. I use data from January 1980 to December 1989 to initialize the model parameters.

State-level Regime Switching. There is substantial state-specific variation in the timing and duration of economic contractions. To highlight the difference in the timing, Figure IA.1 shows the diffusion of the 2007 to 2009 national recession across the 48 contiguous states.[11] The left panel shows that six states were already in a recession three quarters before the nation as a whole went into one (see the dark blue shaded areas). These include states hit particularly hard by the collapse of the housing market (e.g., California and Florida) or by the significant contraction observed in the manufacturing sector (e.g., Ohio). At the national peak, all 48 contiguous states were in a recession, as shown in the middle panel. Finally, the right panel shows that three quarters after the national trough, many states remained in recessionary conditions.[12]

There are also significant cross-sectional differences in the expected duration of state-level expansions and recessions. Table IA.XI reports the expected duration of expansions and recessions. The average expected duration of a recession (expansion) is 12 months (51 months), with a standard deviation of six months (19 months). The table also reports the probability of remaining in each regime. The average probability of remaining in recession and expansion is 97% and 90%, respectively. These numbers suggest that these economic regimes are persistent. Table IA.XI also reports the percentage of time that state-level business cycles are in sync with the national cycle. Following Harding and Pagan (2002), I compute the degree of concordance as

$$C_{i,\mathrm{US}} = \frac{1}{T}\sum_{t=1}^{T}\left[S_{it}S_{\mathrm{US},t} + (1-S_{it})(1-S_{\mathrm{US},t})\right], \tag{IA.10}$$

[11] I say that a U.S. state is in a recession whenever its recession probability is above 50%.

[12] Similar results can be found for the other four national recessions in my sample. The dates for these recessions are 1980.01 to 1980.07, 1981.07 to 1982.11, 1990.07 to 1991.03, and 2001.03 to 2001.11.

where S_{it} and $S_{\mathrm{US},t}$ denote the corresponding economic states. The average state was in the same regime as the country 80% of the time. The standard deviation is equal to 5%.

Table IA.XI.
State-Level Expansions and Recessions

This table reports the expected duration and the probability of remaining in each regime for each state. Following Harding and Pagan (2002), the degree of concordance is given by $C_{i,\mathrm{US}} = \frac{1}{T}\sum_{t=1}^{T}\left[S_{it}S_{\mathrm{US},t} + (1 - S_{it})(1 - S_{\mathrm{US},t})\right]$, where S_{it} and $S_{\mathrm{US},t}$ denote the corresponding economic states. The null column measures the concordance between state i and the U.S. under the null hypothesis that the business cycles of both entities are uncorrelated. The estimation sample runs from January 1980 to December 2019.

State	Expansions Prob. Remaining	Expansions Expected Duration	Recessions Prob. Remaining	Recessions Expected Duration	Concordance	Concordance Null	State	Expansions Prob. Remaining	Expansions Expected Duration	Recessions Prob. Remaining	Recessions Expected Duration	Concordance	Concordance Null
United States	0.98	42.79	0.84	6.36	1.00	0.83	–	–	–	–	–	–	–
Alabama	0.98	41.20	0.91	10.84	0.79	0.74	Montana	0.99	133.29	0.97	34.29	0.79	0.71
Alaska	0.75	4.04	0.97	36.62	0.14	0.14	Nebraska	0.97	35.30	0.89	9.11	0.82	0.73
Arizona	0.98	48.69	0.89	9.40	0.85	0.77	Nevada	0.98	53.73	0.88	8.18	0.91	0.84
Arkansas	0.98	57.92	0.92	12.43	0.86	0.78	New Hampshire	0.99	70.64	0.91	11.23	0.87	0.81
California	0.98	56.09	0.93	14.30	0.85	0.75	New Jersey	0.98	57.00	0.91	11.67	0.87	0.79
Colorado	0.98	57.87	0.93	13.92	0.83	0.77	New Mexico	0.99	91.07	0.92	12.20	0.92	0.83
Connecticut	0.98	41.20	0.93	15.17	0.77	0.68	New York	0.98	62.82	0.92	13.28	0.86	0.78
Delaware	0.96	23.61	0.87	7.41	0.81	0.76	North Carolina	0.98	59.07	0.93	14.40	0.90	0.80
Florida	0.99	79.88	0.97	29.49	0.79	0.70	North Dakota	0.97	30.38	0.84	6.24	0.86	0.80
Georgia	0.98	55.51	0.93	14.80	0.87	0.77	Ohio	0.97	32.89	0.84	6.08	0.88	0.81
Hawaii	0.97	39.43	0.95	19.33	0.71	0.65	Oklahoma	0.97	36.34	0.86	7.38	0.85	0.77
Idaho	0.98	60.38	0.89	8.87	0.90	0.83	Oregon	0.98	46.61	0.90	9.77	0.90	0.79
Illinois	0.98	44.80	0.91	11.23	0.82	0.75	Pennsylvania	0.98	46.40	0.90	10.09	0.84	0.74
Indiana	0.98	40.76	0.92	11.77	0.83	0.74	Rhode Island	0.98	41.58	0.91	11.72	0.79	0.73
Iowa	0.98	50.26	0.90	10.15	0.88	0.80	South Carolina	0.99	68.94	0.88	8.43	0.92	0.83
Kansas	0.98	62.23	0.89	9.23	0.91	0.83	South Dakota	0.97	30.65	0.91	11.12	0.76	0.69
Kentucky	0.98	43.67	0.92	12.16	0.81	0.74	Tennessee	0.98	55.69	0.90	9.95	0.89	0.82
Louisiana	0.99	69.24	0.93	14.21	0.86	0.82	Texas	0.99	73.70	0.93	15.32	0.83	0.79
Maine	0.98	40.06	0.92	12.58	0.82	0.75	Utah	0.99	94.18	0.92	13.05	0.78	0.76
Maryland	0.98	48.25	0.89	8.90	0.84	0.79	Vermont	0.98	46.07	0.88	8.01	0.89	0.81
Massachusetts	0.98	52.76	0.95	19.61	0.76	0.69	Virginia	0.98	57.80	0.93	14.49	0.88	0.79
Michigan	0.97	37.34	0.89	8.91	0.85	0.77	Washington	0.98	44.39	0.89	8.80	0.87	0.78
Minnesota	0.98	43.23	0.91	11.30	0.85	0.77	West Virginia	0.96	25.74	0.89	9.11	0.79	0.70
Mississippi	0.98	46.17	0.91	11.04	0.87	0.78	Wisconsin	0.98	56.45	0.88	8.63	0.92	0.83
Missouri	0.98	44.77	0.90	10.06	0.83	0.77	Wyoming	0.98	64.53	0.84	6.43	0.90	0.86

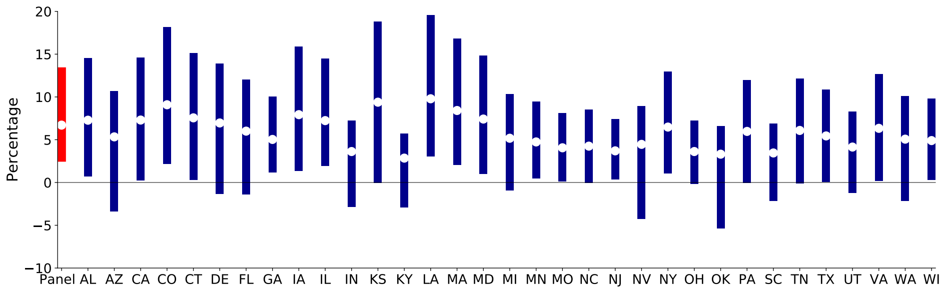

Figure IA.2. State-level unconditional alphas. For each U.S. state, I run a time-series regression of the managed excess state return on the nonmanaged excess state return, $r^{e,\mu}_{j,t+1} = \alpha_j + \beta_j r^{e}_{j,t+1} + \epsilon_{j,t+1}$. The white dots denote OLS point estimates, while the blue shaded bars denote 95% confidence intervals. The red shaded bar denotes estimates from a pooled OLS regression that double–cluster standard errors at the month-state level. To facilitate interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1,200. All models are estimated out of sample on recursively increasing samples of data. I exclude states with less than 10 firms. The forecast evaluation period runs from January 1990 to December 2019.

State-Level Unconditional Alphas. I take the individual states as the unit of analysis and construct regional-based trading strategies by adjusting the conditional beta, $c_j\ \hat{\mu}_{j,t+1|t}$:

$$r^{e,\mu}_{j,t+1} = c_j\ \hat{\mu}_{j,t+1|t}\ r^{e}_{j,t+1}. \tag{IA.11}$$

The subscript j indexes U.S. states, $\hat{\mu}_{j,t+1|t}$ denotes the state j excess return forecast for time $t+1$ using data up to date t, and $r^{e}_{j,t+1}$ is the excess return for state j. Similarly, c_j is a normalization constant equalizing the volatility of $r^{e,\mu}_{j,t+1}$ and $r^{e}_{j,t+1}$.

Figure IA.2 plots the unconditional alphas for each U.S. state in my sample. The white dots denote OLS point estimates, while the blue shaded bars denote 95% confidence intervals. Consistent with the main finding, all point estimates are positive and statistically significant in most cases. This finding is also robust to including the state-level time-series momentum factor and the volatility-managed factor as controls. Finally, the red shaded bar denotes the 95% confidence interval on the point estimate from a pooled OLS regression. To determine this, I stack all states and dates and run a pooled panel regression and double–cluster the standard errors at the month-state level. The panel regression specification utilizes both time-series and cross-sectional variation in state-managed excess returns and state excess returns, increasing the statistical tests' power. The pooled annualized alpha is equal to 6.7%, with a t-statistic of 2.6.

To what extent do the state-level unconditional alphas in Figure IA.2 come from comovement with the national expected return factor versus state-specific factors? To answer this question, each month I sort U.S. states into five buckets based on the predicted state-level expected excess return measure $\hat{\mu}_{j,t+1|t}$. For each bucket, I then compute equally weighted portfolios based on the state portfolios. Finally, to difference out common national factors, I take the difference in returns between the fifth and the first portfolio. Table IA.XII reports the results. The long-short return is economically large and statistically significant. The long portfolio earns a annualized excess return of 11.3%, whereas the short portfolio gains 6.36%. The difference in returns equals 4.95% (t-statistic = 1.9). The mean excess returns of the other three portfolios lie between the mean excess return of the short and long portfolios. Similar results hold true when I compute CAPM alphas.

Overall, this result suggests that the state-level analysis is not driven

Table IA.XII.
State-Level Portfolio Sorts

This table reports performance estimates of a trading strategy that uses the cross-section of state-level expected excess return measures. Each month, I sort U.S. states into five portfolios based on the predicted state-level expected excess return measure $\hat{\mu}_{j,t+1|t}$. The first (fifth) portfolio is the equally weighted portfolio of the 20% of state portfolios with the lowest (highest) expected return measure. The long-short portfolio is the portfolio that captures the difference in the returns of the fifth and the first portfolio. To facilitate interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1,200. All models are estimated out of sample on recursively increasing samples of data. I exclude states with less than 10 firms. The forecast evaluation period runs from January 1990 to December 2019.

	Portfolios					
Return statistics	1	2	3	4	5	long - short
Mean excess return	6.36	8.41	8.88	9.35	11.30	4.95
	[1.46]	[2.32]	[2.88]	[2.66]	[2.45]	[1.91]
α - Mkt	-2.79	-1.02	0.63	1.72	2.70	5.50
	[-1.86]	[-0.62]	[0.48]	[0.72]	[1.95]	[1.82]

entirely by common national variations and provides incremental information for robustness.

J. Timing Volatility

In this section, I combine the information in time-varying expected returns and volatility. Specifically, I modify the conditional beta in equation (**??**) by scaling the expected return by the inverse of the conditional variance.

Table IA.XIII.
Performance Evaluation: Timing Volatility

I run time-series regressions of various trading strategies on the market excess return $r^{e,\mu}_{t+1} = \alpha + \beta r^{e}_{t+1} + \epsilon_{t+1}$. The strategies adjust the conditional beta based on the estimate of conditional expected excess returns, $\hat{\mu}_{t+1|t}$, described in the main text and an estimate of conditional volatility σ^2_t. For σ^2_t, I use the previous month's realized variance. Standard errors adjust for heteroskedasticity and autocorrelation. I report t-statistics in brackets. To facilitate interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1,200. The forecast evaluation period runs from January 1980 to December 2019.

	Conditional beta		
	$c\ \hat{\mu}_{t+1\|t}$	$c\ \frac{1}{\sigma^2_t}$	$c\ \frac{\hat{\mu}_{t+1\|t}}{\sigma^2_t}$
	ERF	Vol	ERF + Vol
	(1)	(2)	(3)
alpha-mkt	5.46	3.51	5.30
	[2.21]	[1.71]	[2.71]

The managed portfolio return is then

$$r^{e,\mu-\sigma}_{t+1} = c\ \frac{\hat{\mu}_{t+1|t}}{\sigma^2_t} \cdot r^{e}_{t+1}, \tag{IA.12}$$

where I again set c such that $r^{e,\mu-\sigma}_{t+1}$ has the same unconditional volatility as r^{e}_{t+1}. The main motivation for the use of these weights comes from the optimal dynamic strategy of a mean-variance investor who levers their position up or down on the risky asset over time to maximize the unconditional Sharpe ratio of the portfolio (see Daniel and Moskowitz (2016)). To compute the weights, I use the main specification for $\hat{\mu}_{t+1|t}$. For σ^2_t, I follow Moreira and Muir (2017) and use the previous month's realized variance, which is computed from the daily data. Specifically, I compute σ^2_t using

$\sigma_t^2 = \sum_{d=1}^{T_t} = (r_{d,t}^e - \frac{\sum_{d=1}^{T_t,t} r_{d,t}^e}{T_t})^2$, where t denotes the month and T_d the total number of days in month t.

Table IA.XIII reports the results. Column (2) shows that a strategy that adjusts the conditional beta using the inverse of the conditional volatility (i.e., $\frac{c}{\sigma_t^2}$) generates an alpha equal to 3.51% (t-statistic = 1.71). Column (3) shows that if I consider both expected returns and volatility, the alpha equals 5.30% (t-statistic = 2.71).

J.1. Sorts on the One-Month-Ahead Excess Return Forecasts

For the aggregate market and U.S. state portfolios, I bucket the period t excess return forecast (using only data up until time $t-1$) into 10 bins (low to high). For each bin, I then report average values and 95% confidence intervals for the average conditional risk exposure and the trading strategy's excess return over the subsequent month. The top, middle, and bottom panels use as conditional beta $c\,\hat{\mu}_{t+1|t}$, $c\,\frac{1}{\sigma_t^2}$, and $c\,\frac{\hat{\mu}_{t+1|t}}{\sigma_t^2}$, respectively.

Figure IA.3 shows that all strategies take substantially less risk when the excess return forecast (ERF) is in the first decile. Note that at such times, the conditional risk exposure is negative or close to zero, and the trading strategies' excess returns are positive or zero but not negative. The figure further shows that when the ERF is in the last decile, the volatility-timing strategies have relatively low risk exposure. At such times, volatility is high, and, as shown in Figure 6 in the main text, the probability of being in a recession is also high. Therefore, volatility-timing strategies stay out of the market for too long and miss subsequent higher-than-average excess returns.

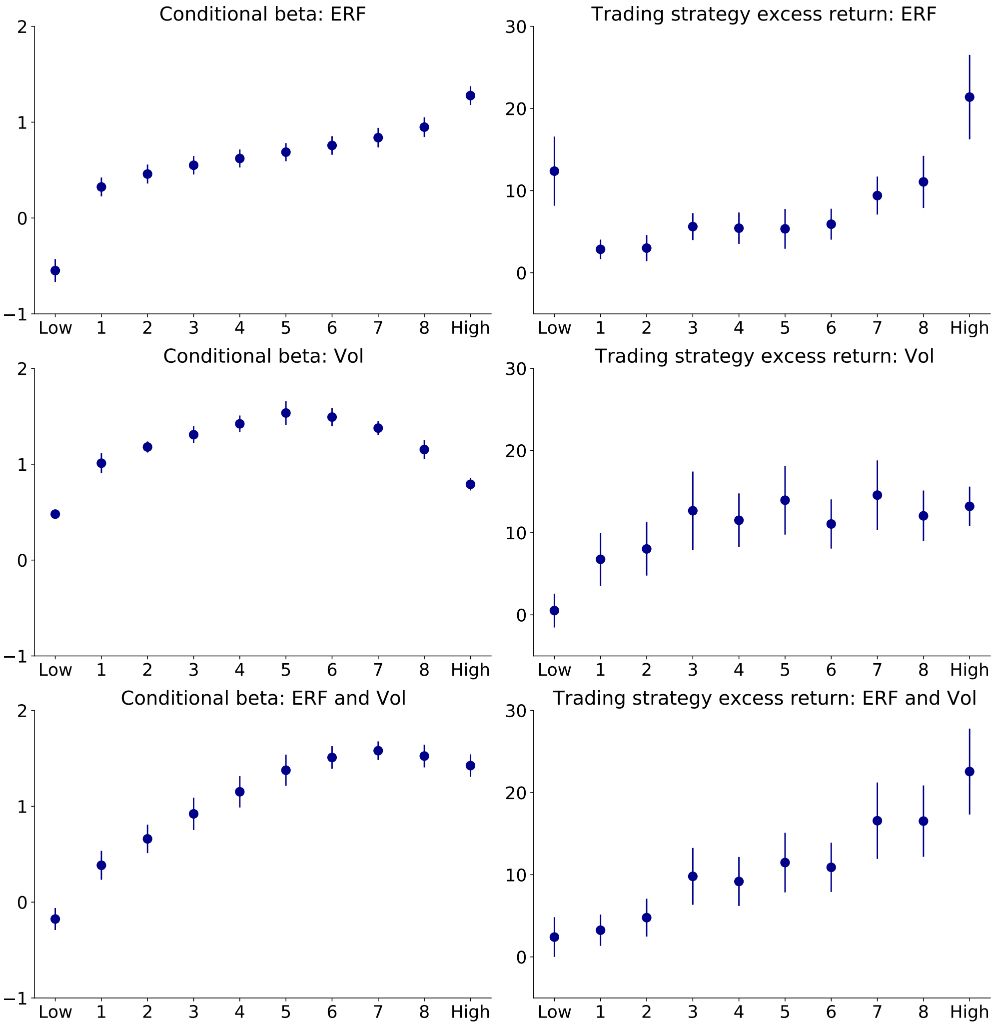

Figure IA.3. Sorts on the one-month-ahead excess return forecasts. This figure sorts the period t excess return forecast (ERF) for the aggregate market and U.S. state portfolios into 10 buckets. For each decile, the figure reports average values and 95% confidence intervals for the average conditional risk exposure and the trading strategies' excess returns over the subsequent month. The top, middle, and bottom panels use as conditional beta $c\ \hat{\mu}_{t+1|t}$, $c\ \frac{1}{\sigma_t^2}$, and $c\ \frac{\hat{\mu}_{t+1|t}}{\sigma_t^2}$, respectively. These estimates use data up to time $t-1$.

J.2. Spanning Test

In this section, I conduct formal spanning tests among these three different portfolios. Specifically, I consider the strategies that adjust risk exposure based on the following three conditional beta estimates: $c\,\hat{\mu}_{t+1|t}$ (ERF), $c\,\frac{1}{\sigma_t^2}$ (Vol), and $c\,\frac{\hat{\mu}_{t+1|t}}{\sigma_t^2}$ (ERF+Vol). I regress these portfolio returns on each other. The annualized alphas from these regressions are reported in Table IA.XIV. For ease of comparison, the three strategies have the same volatility.

Panel A of Table IA.XIV reports results from regressing the Vol and ERF+Vol portfolio's returns on the ERF return. The intercepts are positive and significant in most cases, with t-statistics ranging from 1.46 to 3.57. These results indicate that the ERF strategy does not capture the Vol and ERF+Vol strategies. Panel B of Table IA.XIV regresses the returns of the ERF and ERF+Vol strategies on the volatility-managed factor, Vol. All of the alphas are positive and highly statistically significant, suggesting that the ERF and ERF+Vol strategies are not spanned by the volatility-managed factor. Finally, Panel C examines whether the ERF and Vol portfolios are spanned by the ERF+Vol portfolio. Columns (1) and (4) indicate that the ERF strategy is not spanned by the ERF + Vol portfolio. However, the alphas for the Vol strategy are small and statistically indistinguishable from zero. This result suggests that ERF + Vol portfolio spans the Vol portfolio.

K. Difference in Cumulative Squared Prediction Error

Figure IA.4 shows the cumulative squared prediction errors of the benchmark model minus the cumulative squared prediction error of $\hat{\mu}_{t+1|t}$ (see

Table IA.XIV.
Spanning Tests

This table presents results of spanning tests. I consider three different strategies that adjust risk exposure based on the following three conditional beta estimates: $c\ \hat{\mu}_{t+1|t}$ (ERF), $c\ \frac{1}{\sigma_t^2}$ (Vol), and $c\ \frac{\hat{\mu}_{t+1|t}}{\sigma_t^2}$ (ERF+Vol). Panel A regresses Vol and ERF+Vol on ERF. Panel B regresses ERF and ERF+Vol on Vol. Panel C regresses ERF and Vol on ERF+Vol. For ease of comparison, the three strategies have the same volatility. I report annualized alphas from these regressions.

				Cross-sectional evidence: U.S. States		
	ERF	Vol	ERF + Vol	ERF	Vol	ERF + Vol
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Independent variable = returns to the strategy that uses $c\ \hat{\mu}_{t+1\|t}$ (ERF)						
α	–	3.30	3.39	–	7.94	6.43
	–	[1.46]	[1.91]	–	[3.31]	[3.57]
R-squared	–	0.33	0.48	–	0.10	0.28
Observations	–	479	479	–	11847	11847
Panel B: Independent variable = returns to the strategy that uses $c\ \frac{1}{\sigma_t^2}$ (Vol)						
α	5.05	–	2.10	4.90	–	2.61
	[2.99]	–	[2.17]	[2.25]	–	[2.76]
R-squared	0.33	–	0.83	0.10	–	0.60
Observations	479	–	479	11847	–	11847
Panel C: Independent variable = returns to the strategy that uses $c\ \frac{\hat{\mu}_{t+1\|t}}{\sigma_t^2}$ (ERF + Vol)						
α	3.07	-0.40	–	2.51	2.16	–
	[2.34]	[-0.39]	–	[1.92]	[1.43]	–
R-squared	0.48	0.83	–	0.28	0.60	–
Observations	479	479	–	11847	11847	–

Section II in the main article for details on how to compute $\hat{\mu}_{t+1|t}$). I take as a benchmark the return forecast of a model that uses equations (2) and (3) in the main text but sets μ_t equal to a constant, that is, $r^e_{t+1} = \mu_0 + \epsilon^r_{t+1}$. The figure shows that the slope of the cumulative difference in squared prediction errors is mostly positive over the evaluation period. This result suggests that the predictive power of $\hat{\mu}_{t+1|t}$ is not concentrated in relatively short-lived

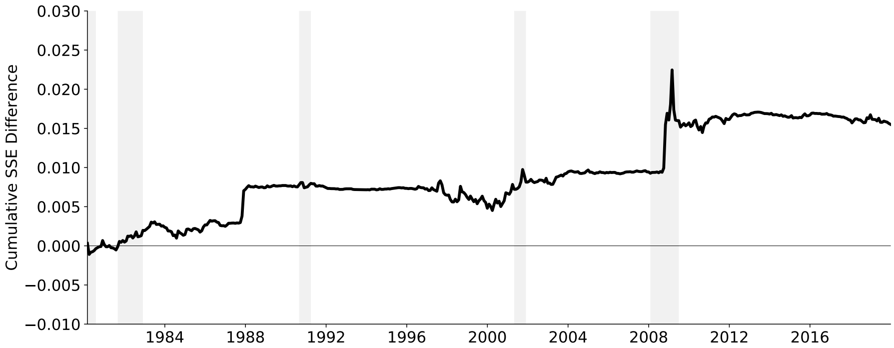

Figure IA.4. Difference in cumulative squared prediction error. This figure shows the cumulative squared prediction errors of the benchmark model minus the cumulative squared prediction error of the main econometric framework described in Section II in the main article. The benchmark model is the return forecast of a model that uses equations (2) and (3) in the main text but sets μ_t equal to a constant, that is, $r^e_{t+1} = \mu_0 + \epsilon^r_{t+1}$. The forecast evaluation period runs from January 1980 to December 2019.

periods or predictability pockets.

L. Beta During Expansions

This section analyzes the behavior of expected returns across the business cycle. I run contemporaneous time-series regressions of the strategy's risk exposure (i.e., $c\ \hat{\mu}_{t+1|t}$) onto five variables related to future stock market returns as previously documented in the literature.

The first variable that I consider is investors' expectations of future stock market returns, obtained from the American Association of Individual Investors (AAII) survey. Each week since 1987, the AAII has asked its members a simple question: "Do you feel the direction of the stock market over the next six months will be up (bullish), no change (neutral) or down (bearish)?" My measure of expectations is the monthly average of the percentage of "bullish" investors minus the percentage of "bearish" investors (i.e., %Bull - %Bear). Greenwood and Shleifer (2014) also rely on the AAII survey data. My second variable is the investor sentiment measure of Baker and Wurgler (2006). My third variable is monthly investor inflows into equity-oriented mutual funds from the Investment Company Institute. In each period, I scale the inflows by the aggregate capitalization of the U.S. stock market. The fourth variable is the year-on-year change in the Coincident Index, which measures the current state of the business cycle. The fifth and last variable is the dividend-price ratio, which is a typical measure of risk premia in the literature. I standardize these variables to ease the interpretation of the regression coefficients.

Table IA.XV presents the results. As shown in columns (1), (2), and (3), during NBER expansions, when the survey measure of return expectations, sentiment, or flows into the equity markets increase, the strategy takes a contrarian position and reduces risk-taking.[13] In good times, these results are broadly consistent with studies in the behavioral literature suggesting that many investors hold extrapolative expectations about returns. After a period of good stock market performance, one that generates high sentiment, many investors become more bullish. Therefore, *extrapolators* rebalance their

[13]The negative relation between the estimate of expected returns and the survey measures of return expectations in expansions is consistent with the findings of Greenwood and Shleifer (2014), who show that six investor surveys (including the AAII survey) are negatively correlated with traditional predictors of returns, such as the dividend-price ratio.

portfolios into equity, and, doing so, bid up current stock prices even further. At this point, the stock market is arguably overvalued, and hence subsequent price changes are low, on average. In essence, the strategy exploits this mispricing by mildly decreasing the portfolio weights on stocks. For instance, see from Figure 3 that during expansions the strategy negatively weights lag returns to predict future returns. This is by and large consistent with the equilibrium model of Barberis et al. (2015) in which rational traders counteract the overvaluation caused by extrapolators. Therefore, my expected return estimates can be conceptualized as reflecting the return expectations of rational traders in their model. In contrast, extrapolation of returns does not explain my strategy's profits during recessions. Note that during these bad economic times the strategy follows the survey and sentiment measures as well as the flows into equity markets. However, for the former two series, the point estimates are close to zero.

During expansions, the strategy's profits can also be explained in terms of risk. Columns (4) and (5) show that during expansions, when economic conditions deteriorate (but not enough to generate a recession) or when prices decrease more than dividends (representing an increase in risk premia), the strategy increases risk-taking. These results imply that during expansions, the strategy has a strong countercyclical component. Hence, profits also can be viewed as compensation for bearing risk.[14]

[14] For example, in a standard representative-agent asset pricing model, risk premia vary over the business cycle and are relatively high in bad times when the economy is slowing down. For habit models, see Campbell and Cochrane (1999). For long-run risk models, see Bansal and Yaron (2004) and Bansal et al. (2012). For models with rare disasters, see

While I cannot determine whether the strategy's profits during expansions are due to market inefficiency or to rational variation in expected returns, my profits during recessions conflict with standard risk-based explanations. Specifically, during recessionary periods, an increase in the dividend-price ratio or a decrease in macro growth would suggest high expected returns and thus a time to buy stocks. Yet my strategy rebalanced the portfolio weights out of equity and into Treasuries.

Rietz (1988), Barro (2006), Gabaix (2012), and Wachter (2013).

Table IA.XV.
Regression of Risk Exposure on Contemporaneous Variables

I run time-series regressions of the strategy's risk exposure on various contemporaneous variables $c\,\hat{\mu}_{t+1|t} = (\omega_0 + \omega_{rec} \times I_{rec,t}) + (\beta_0 + \beta_{rec} \times I_{rec,t})\, x_t + \epsilon_t$, where $I_{rec,t}$ denotes the NBER recessionary index and x_t denotes one of the contemporaneous variables considered. The data are monthly on the sample period 1987 to 2016 given that the %Bull - %Bear series from the American Association of Individual Investors survey starts in 1987. The sentiment measure of Baker and Wurgler (2006) ends in 2015:M9. Standard errors are adjusted for heteroskedasticity and autocorrelation. t-statistics are reported in brackets.

	Dependent variable: $c\,\hat{\mu}_{t+1\|t}$					
Contemporaneous variables	(1)	(2)	(3)	(4)	(5)	(6)
%Bull - %Bear	-0.32					-0.14
	[-2.57]					[-1.47]
%Bull - %Bear $\times I_{rec}$	0.36					-0.08
	[1.89]					[-0.27]
Sentiment		-0.31				-0.14
		[-4.21]				[-2.18]
Sentiment $\times I_{rec}$		0.33				-0.09
		[2.44]				[-0.46]
Flow into equity markets			-0.17			-0.07
			[-2.51]			[-1.32]
Flow into equity markets $\times I_{rec}$			0.75			0.67
			[7.25]			[6.80]
Coincident indicator YoY				-0.33		-0.25
				[-4.51]		[-3.62]
Coincident indicator YoY $\times I_{rec}$				0.56		0.58
				[2.49]		[3.84]
Dividend-price ratio					0.24	0.15
					[2.82]	[1.54]
Dividend-price ratio $\times I_{rec}$					-0.34	-0.53
					[-1.41]	[-2.47]
ω_0	0.79	0.77	0.77	0.81	0.77	0.84
	[9.84]	[8.65]	[8.41]	[10.32]	[8.80]	[14.06]
$\omega_{rec} \times I_{rec}$	-0.38	-0.41	0.03	-0.14	-0.36	0.37
	[-2.15]	[-1.38]	[0.15]	[-0.47]	[-1.66]	[0.79]
R^2-adjusted	0.19	0.17	0.20	0.22	0.14	0.54

II. Forecasting Model: Data, Details, and Proofs

The data set consists of macroeconomic variables measured at the monthly and quarterly frequencies. All of the series start at different periods, but all end in December of 2019.

A. *State-Space Representations*

In this section, I describe the state-space representation of the forecasting model.

A.1. State-Space Representation of the Forecasting Model Weights

Below, I describe the state-space representation of the forecasting model weights. The representation consists of measurement equation

$$y_{t+1} = A_{t+1}\left(H_0 + H_1 x_{t+1} + \nu_{t+1}\right) \qquad \text{with} \qquad \nu_{t+1} \sim N(0, R); \tag{IA.13}$$

where A_{t+1} is a selection matrix that accounts for changes in the vector of observables y_{t+1}. The state-transition equation is

$$x_{t+1} = F_0(S_t, S_{t+1}) + F_1(S_{t+1})x_t + \omega_{t+1} \qquad \text{with} \qquad \omega_{t+1} \sim N(0, Q(S_{t+1})). \tag{IA.14}$$

I will describe each in turn.

Measurement Equation. The measurement equation provides the link be-

Table IA.XVI.
Data Series Underlying the Aggregate Economic Activity Index

The first column reports the macro series used to compute the monthly measure of economic activity. The second column denotes the source. The abbreviations are as follows: BLS = U.S. Bureau of Labor Statistics, BEA = U.S. Bureau of Economic Analysis, and ETA = U.S. Employment and Training Administration. The third column reports the frequency. The fourth column gives the first observation available. All series end in December 2019.

Data series (1)	Source (2)	Frequency (3)	First obs. (4)
Industrial production index	BEA	Monthly	1962:M10
Nonfarm payroll employment	BLS	Monthly	1964:M11
Real personal income excluding current transfer receipts	BEA	Monthly	1959:M2
Aggregate weekly hours worked	BLS	Monthly	1971:M8
Initial jobless claims	ETA	Monthly	1967:M2
GDP	BEA	Quarterly	1965:Q3
Personal consumption expenditures	BEA	Quarterly	1965:Q4
Private residential fixed investment	BEA	Quarterly	1965:Q4
Private nonresidential fixed investment	BEA	Quarterly	1965:Q4

Table IA.XVII.
Data Series Underlying the U.S. State Economic Activity Index

The first column reports the macro series used to compute the monthly measure of economic activity. The second column denotes the source. The abbreviations are as follows: BLS = U.S. Bureau of Labor Statistics, BEA = U.S. Bureau of Economic Analysis, ETA = U.S. Employment and Training Administration., FHFA = U.S. Federal Housing Finance Agency. The third column reports the frequency. The fourth column gives the first observation available. All series end in December 2019.

Data series (1)	Source (2)	Frequency (3)	First obs. (4)
Nonagricultural payroll employment	BLS	Monthly	1990:M1.
Unemployment rate	BLS	Monthly	1976:M1
Initial jobless claims	ETA	Monthly	1986:M2
Real wage and salary disbursements	BEA	Quarterly	1998:Q3
Personal income	BEA	Quarterly	1948:Q3
House price index	FHFA	Quarterly	1975:Q3

tween the vector of observables y_{t+1} and the vector of state variables, x_{t+1}. This link is determined by the assumed processes in equation (??),

$$y_{i,t+1} = \gamma_i z_{t+1} + e_{i,t+1} \quad \text{for} \qquad i = 1, ..., N_{\text{Macro}}. \tag{IA.15}$$

To facilitate model estimation, I further specify $y^*_{i,t+1} = y_{i,t+1} - \psi_i y_{i,t}$. Hence, equation (IA.15) simplifies to

$$y^*_{i,t+1} = \gamma_i z_{t+1} - \gamma_i \psi_i z_t + \sigma_i \epsilon_{i,t+1} \quad \text{for} \qquad i = 1, ..., N_{\text{Macro}}.$$

Conditional on the parameters, the vector of observables is given by $y_{t+1} = [y^*_{1,t+1}, y^{*,qrt}_{2,t+1}]'$. For the sake of exposition, I consider only two observables (i.e., $N_{\text{Macro}} = 2$), and I further assume that the first variable is available at the monthly frequency (e.g., industrial production growth), while the second variable is available only at the quarterly frequency (e.g., GDP growth). The link between the quarterly time-aggregated series with the model-implied monthly frequency is given by the following function described in the main article:

$$y^{*,qrt}_{2,t+1} = \sum_{j=1}^{5} \frac{3 - |j-3|}{3} y_{2,t+2-j}, \tag{IA.16}$$

such that the monthly model variables $y_{2,t+2-j}$ time-aggregate to the observed quarterly growth rates $y^{*,qrt}_{2,t+1}$ (to simplify notation, I further omit the o subscript used for the observable variables in the main text).

The selection matrix A_{t+1} and the vector of observables y_{t+1} are time dependent. Specifically, we have:

- If $t+1$ is the last month of the quarter:

$$y_{t+1} = \begin{bmatrix} y^*_{1,t+1} \\ y^{*,qrt}_{2,t+1} \end{bmatrix}_{2\times 1} \qquad A_{t+1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2}.$$

- If $t+1$ is the first or second month of the quarter:

$$y_{t+1} = \begin{bmatrix} y^*_{1,t+1} \end{bmatrix}_{1\times 1} \qquad A_{t+1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}_{2\times 2}.$$

To define the matrices associated with the measurement equation (i.e., H_0, H_1, and R), I first specify the state vector x_{t+1} as

$$x_{t+1} = \begin{bmatrix} z_{t+1} & z_t & z_{t-1} & z_{t-2} & z_{t-3} & z_{t-4} & \sigma_1\epsilon_{1,t+1} & \sigma_2\epsilon_{2,t} & \sigma_2\epsilon_{2,t-1} & \sigma_2\epsilon_{2,t-2} & \sigma_2\epsilon_{2,t-3} & \sigma_2\epsilon_{2,t-4} \end{bmatrix}'_{12\times 1}.$$

Given the state vector x_{t+1}, it follows that

$$H_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}_{2\times 1} \qquad R = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times 2}$$

$$H_1 = \begin{bmatrix} \gamma_1 & -\gamma_1\psi_1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3}\gamma_2 & \frac{2}{3}\gamma_2 - \frac{1}{3}\gamma_2\psi_2 & \frac{3}{3}\gamma_2 - \frac{2}{3}\gamma_2\psi_2 & \frac{2}{3}\gamma_2 - \frac{3}{3}\gamma_2\psi_2 & \frac{1}{3}\gamma_2 - \frac{2}{3}\gamma_2\psi_2 & -\frac{1}{3}\gamma_2\psi_2 & 0 & \frac{1}{3} & \frac{2}{3} & \frac{3}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}_{2\times 12}.$$

State-Transition Equation. The state-transition equation provides the law of motion of the state variables in the model, which evolve according to

$$z_{t+1} = \mu_z(S_{t+1}) + \phi_z(S_{t+1})\left(z_t - \mu_z(S_t)\right) + \sigma_z(S_{t+1})\epsilon_{z,t+1}, \tag{IA.17}$$

Given x_{t+1}, it follows that

$$F_0(S_t, S_{t+1}) = \begin{bmatrix} \mu_z(S_{t+1}) - \phi_z(S_{t+1})\mu_z(S_t) \\ O_{11\times 1} \end{bmatrix}_{12\times 1} \quad F_1(S_{t+1}) = \left[\begin{array}{cccccc:c} \phi_z(S_{t+1}) & 0 & 0 & 0 & 0 & 0 & \\ 1 & 0 & 0 & 0 & 0 & 0 & \\ 0 & 1 & 0 & 0 & 0 & 0 & \\ 0 & 0 & 1 & 0 & 0 & 0 & O_{6\times 6} \\ 0 & 0 & 0 & 1 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 1 & 0 & \\ \hdashline & & O_{6\times 6} & & & & O_{6\times 6} \end{array}\right]_{12\times 12},$$

and $Q(S_{t+1})$ is a sparse 12×12 matrix with the following elements in the ij^{th} position:

$$Q^{[1,1]}(S_{t+1}) = \sigma^2_{z,S_{t+1}}, \quad Q^{[7,7]}(S_{t+1}) = \sigma^2_1, \qquad Q^{[8,8]}(S_{t+1}) = \sigma^2_2.$$

A.2. State-Space Representation of the Process for the Excess Market Return

Below, I describe the state-space representation of the process for the excess market return. In the main text, I defined

$$\begin{aligned} r^e_{t+1} &= \mu_t + \varepsilon^r_{t+1}, \\ \mu_{t+1} &= \mu_0 + \rho(\mu_t - \mu_0) + \varepsilon^\mu_{t+1}, \end{aligned} \tag{IA.18}$$

with $(\varepsilon^r_{t+1}, \varepsilon^\mu_{t+1})' \sim iidN\left(\underline{0}, \Sigma(S_t)\right)$, where

$$\Sigma(S_t) = \begin{bmatrix} \sigma^2_r(S_t) & \sigma^2_{r\mu}(S_t) \\ \sigma^2_{\mu r}(S_t) & \sigma^2_\mu(S_t) \end{bmatrix}.$$

To write the state-space model, I use the same notation as in equations (IA.13) and (IA.14).

Measurement Equation. For each time period $t+1$, the vector of observables and the selection matrix are $y_{t+1} = r^e_{t+1}$ and $A_{t+1} = 1$, respectively. I write the rest of the matrices as follows:

$$x_{t+1} = \begin{bmatrix} \mu_{t+1} & \mu_t & \sigma_r(S_t)\varepsilon^r_{t+1} \end{bmatrix}'_{3\times 1}, \quad H_0 = \begin{bmatrix} 0 \end{bmatrix}_{1\times 1}, \quad H_1 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}_{1\times 3}, \quad R = \begin{bmatrix} 0 \end{bmatrix}_{1\times 1}.$$

State-Transition Equation. Given the state vector x_{t+1}, the matrices associated with the state-transition equation are given by

$$F_0 = \begin{bmatrix} \mu_0(1-\rho) \\ 0 \\ 0 \end{bmatrix}_{3\times 1}, \quad F_1 = \begin{bmatrix} \rho & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3}, \quad Q(S_t) = \begin{bmatrix} \sigma^2_r(S_t) & 0 & \sigma^2_{r\mu}(S_t) \\ 0 & 0 & 0 \\ \sigma^2_{r\mu}(S_t) & 0 & \sigma^2_\mu(S_t) \end{bmatrix}_{3\times 3}.$$

B. Posterior Inference: Forecasting Model Weights

To estimate the model associated with the forecasting weights described in Section A.1, I use the Bayesian Gibbs sampling approach described in Kim and Nelson (1998).[15] The assumed prior distribution is quite diffuse, and is similar to the one used in Kim et al. (1999). The estimation's main output is the posterior distribution of the economic states S_t. The posterior mean of S_t represents the real-time recessionary probabilities, π_t. Importantly, these

[15]For a textbook description of the Gibbs sampler, see Chapter 10 of Kim et al. (1999).

probabilities correspond to the dynamic forecasting weights of the predictive model described in Section II.

C. Posterior Inference: Process for Excess Returns

The state-space representation described in Section A.2 is linear conditional on the economic states S_t (estimated in the previous stage, Section B). Therefore, for posterior inference of the aggregate predictive model, I can use a standard Metropolis-within-Gibbs sampler.

Algorithm. Let Θ denote the parameter vector and $\mu_{1:T}$, be the sequence of hidden states. The Metropolis-within-Gibbs algorithm involves iteratively sampling from two conditional posterior distributions. For $s = 1, \ldots, N_{sim}$, the main steps of the sampler are:

1. Initialize $\Theta^{(0)}$ at some value.
2. Draw $\mu_{1:T}^{(s)}$ conditional on $\Theta^{(s-1)}$ and the vector of observables $y_{1:T}$. I obtain draws of $\mu_{1:T}^{(s)}$ by applying the simulation smoother of Durbin and Koopman (2012).
3. Draw $\Theta^{(s)}$ conditional on $\mu_{1:T}^{(s)}$ and the vector of observables $y_{1:T}$. I use a Random-Walk Metropolis-Hastings algorithm to generate draws of the model parameters, Θ, from their posterior distribution in (**??**). Furthermore, I assume that the proposal distribution is $N(\Theta^{(s-1)}, c\,\Omega)$ for $s = 1, \cdots, N_{sim}$.

For the covariance matrix of the proposal distribution, Ω, I follow Schorfheide (2005) and use the negative inverse of the posterior density Hessian evaluated at the posterior mode $\tilde{\Theta}$. I target an acceptance rate of approximately 30% by adequately modifying the scaling factor c. I initialize the algorithm by selecting $\Theta^{(0)}$ to be the posterior mode $\tilde{\Theta}$ (estimated using standard maximization algorithms) and generate 25,000 draws (i.e., $N_{sim} = 25,000$) from the posterior distribution. Finally, I set the burn-in period at 15,000 draws.

D. Identification

In this section, I establish the identification results of the predictive model. A simple way to check identification is to rewrite the state-space model into an ARMA(1,1) process, where we know that the unknown model parameters can be consistently estimated provided that the roots of the MA component are normalized to lie on or outside the unit circle, and are different from the roots of the AR component, assuming that these roots also lie outside the unit circle (see Griliches et al. (1983) for instance). In this section, I assume that the ARMA(1,1) model is identified, that is, a change in any parameter associated with the ARMA(1,1) would imply a distinct probability distribution for the aggregate market return $\{r_t^e\}_{t=1}^T$. I then compare the parameters of the ARMA(1,1) model with the parameters implied by the state-space representation.

ARMA(1,1) Representation. To rewrite the model into an ARMA(1,1) process, consider the following state-space model

$$
\begin{aligned}
&r^e_{t+1} = \mu_{t+1} + \varepsilon^r_{t+1}, \\
&\mu_{t+1} = \rho(S_t)\mu_t + \varepsilon^\mu_{t+1}, \\
&(\varepsilon^r_{t+1}, \varepsilon^\mu_{t+1})' \sim iidN(\underline{0}, \Sigma(S_t)) \qquad \text{with} \quad \Sigma(S_t) = \begin{bmatrix} \sigma^2_r(S_t) & \sigma^2_{r\mu}(S_t) \\ \sigma^2_{\mu r}(S_t) & \sigma^2_\mu(S_t) \end{bmatrix},
\end{aligned} \tag{B.1}
$$

where to simplify the subsequent notation I set the constant terms to zero, which can be identified from the conditional mean of r^e_{t+1}. I also slightly modify the timing in the measurement equation to simplify the notation of the ARMA(1,1) process. However, this change in timing does not affect the identification issues of the model that this section addresses.

To reformulate the state-space into an ARMA(1,1) model, we can combine the measurement equation with the state-transition equation as follows:

$$
\begin{aligned}
r^e_{t+1} &= \mu_{t+1} + \varepsilon^r_{t+1}, \\
&= \left(\rho(S_t)\mu_t + \varepsilon^\mu_{t+1}\right) + \varepsilon^r_{t+1} \\
&= \rho(S_t) r^e_t - \rho(S_t)\varepsilon^r_t + \varepsilon^\mu_{t+1} + \varepsilon^r_{t+1}.
\end{aligned}
$$

Since conditional on being in regime S_t, ε^μ_{t+1} and ε^r_{t+1} are normally distributed, their sum, $\eta_{t+1} = \varepsilon^\mu_{t+1} + \varepsilon^r_{t+1}$, follows a normal distribution with zero mean and variance $\sigma^2_\eta(S_t) = \sigma^2_r(S_t) + \sigma^2_\mu(S_t) + 2\sigma^2_{r\mu}(S_t)$. Hence, if we let

$\eta_{t+1} \sim N(0, \sigma_\eta^2(S_t))$ and rescale the variance, it follows that

$$
\begin{aligned}
r_{t+1}^e &= \rho(S_t) r_t^e - \rho(S_t) \frac{\sigma_r(S_t)}{\sqrt{\sigma_r^2(S_t) + \sigma_\mu^2(S_t) + 2\sigma_{r\mu}^2(S_t)}} \eta_t + \eta_{t+1} \\
&= \gamma(S_t) r_t^e + \psi(S_t)\eta_t + \eta_{t+1},
\end{aligned} \tag{B.2}
$$

which is a Markov-switching ARMA(1,1) process with AR coefficient $\gamma(S_t)$ and MA and variance parameters equal to $\psi(S_t)$ and $\sigma_\eta^2(S_t)$, respectively. Hence, the ARMA(1,1) has six free parameters. I impose two parameter restrictions to achieve identification. The first restriction fixes the steady-state variance of r^e conditional on the expansionary regime. The second sets the correlation parameter between the unexpected return and expected return shocks to be equal across regimes. Under these identifying restrictions, the number of model parameters equals the number of free parameters of the ARMA(1,1) representation and therefore the model is identified.

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